

Factors associated with discontinued participation in a citywide fruit and vegetable incentive program: The role of community-based organizations in engaging non-English speakers

ABSTRACT

Nutrition incentive programs are an important fiscal tool to address inequities in healthy food consumption. Fresh Bucks provides US\$40/month for fruit and vegetable (FV) purchases to Seattle low-income households, allowing long-term participation, and partnering with community-based organizations (CBOs) to enroll non-English speakers. Previous research shows Fresh Bucks participation reduces food insecurity and increases FV consumption. We aimed to identify factors associated with program discontinuation ~2 years post-initial enrollment, particularly among non-English speakers. We used administrative and survey data from enrollment application (Jan-Mar 2020) and re-application (Oct-Dec 2021) to identify program discontinuation (n=12,813). A linear probability model included baseline age, race/ethnicity, preferred language, enrollment pathway (public vs. CBO-assisted), FV consumption (<3 times/day, ≥ 3 times/day, or missing), household income, presence of children, and food security. We ran pooled and language-stratified analyses. Rates of program discontinuation were 46% (total), 37.6% (English speakers), and 53.3% (non-English speakers). In pooled analyses, age ≥ 30 years (vs. 18-29); having children; income-to-poverty ratio (IPR)=100-200% or IPR > 200% (vs. IPR < 100%); being non-Hispanic (NH) Asian (vs. NH White); speaking Chinese, Spanish (vs. English); and being food secure were associated with lower probability of program discontinuation. Being NH Black (vs. NH White) was associated with higher probability of program discontinuation. CBO-assisted enrollment (vs. public) was associated with lower

probability of discontinuation for non-English speakers (-7.3 percentage points, 95% confidence interval= -10.5, -4.1). In conclusion, age, income, race/ethnicity, preferred language, income, and food security predicted program discontinuation. Community engagement was important for retaining non-English speakers, offering insights for improving retention in nutrition incentive programs.

Keywords: fruits; vegetables; incentive program; community-engagement; language.

Introduction

Fruits and vegetables are critical components of a healthy diet, yet their consumption in the United States (U.S.) is low. In 2019, 12.3% and 10% of U.S. adults met recommendations for fruit and vegetable consumption, respectively, with significant variations by sociodemographic characteristics (Lee et al., 2022). Only 6.8% of U.S. adults in low-income households met vegetable recommendations, compared to 12.2% of adults in the highest income category (Lee et al., 2022). This disparity is partly explained by the higher cost of fruit and vegetables, relative to other more energy-dense foods (Yeh et al., 2008; Darmon & Drewnowski, 2015). A public health strategy designed to address this cost barrier are nutrition incentive programs, which provide low-income households with either dollar-to-dollar match incentives or direct funds for the purchase of healthy foods.

Fresh Bucks is one such program, providing low-income households in Seattle, WA with \$40/month to purchase fruits and vegetables in participating food outlets (City of Seattle, n.d.). Launched in 2020, Fresh Bucks allowed eligible Seattle residents to apply for the program via their public website. In addition, Fresh Bucks partnered with community-based organizations (CBOs) in the city to assist with enrolling and engaging with traditionally underserved groups, including non-English speakers. Previous research found that Fresh Bucks participation reduced food insecurity and increased fruit and vegetable consumption (Knox et al., 2025). Further, research showed that redemption of Fresh Bucks benefits was higher among those who preferred to speak Chinese, Vietnamese, or Spanish, compared to those who preferred to speak English (Wallace et al., unpublished results (a)).

Unlike other nutrition incentive programs which may have limited durations, Fresh Bucks allows participants to stay in the program long-term. However, due to post-pandemic fiscal

changes, Fresh Bucks requested participants to re-apply for the program ~2 years after launching. The aims of this study were to identify factors associated with discontinuation of Fresh Bucks participation 2 years after initial launch, and to assess if these factors vary by participants' preferred language. These research questions have important programmatic implications, informing the retention efforts of Fresh Bucks and other similar nutrition incentive programs, particularly for historically underserved groups.

Methods

We used data from administrative records and surveys collected by the Fresh Bucks program at the time of first program enrollment (January-March 2020) and when participants were required to re-apply to continue to receive benefits (July-December 2021, to maintain benefits after January 1, 2022). Fresh Bucks staff linked administrative and survey data through a household ID. They then shared the linked data with the researchers without including personally identifiable information. We used the re-application administrative data to identify participants who discontinued the program (i.e., those who did not re-apply), with program discontinuation being the dependent variable of interest. We used the baseline administrative and survey data to extract independent variables, as described below. We had data from all sources from 12,813 participants.

Baseline sociodemographic information from Fresh Bucks participants obtained from administrative data included participants' age, race/ethnicity (non-Hispanic [NH] Asian, NH Black, Hispanic/Latino, NH White, NH Other), preferred language (Chinese, English, Spanish, Vietnamese, Another non-English language), having a child in the household (yes/no), household

size, and household income. We used household income and size, plus the 2021 federal poverty guidelines, to calculate participants' household income-to-poverty ratio (IPR). We also obtained information on participant's Fresh Bucks enrollment pathway, categorized as public (i.e., participants signed up via the Fresh Bucks website) or CBO-assisted (i.e., participants signed up with assistance from one of the Fresh Bucks affiliated CBOs).

From the baseline survey, we obtained information on household food security status and participants' fruit and vegetable consumption. Household food security was assessed with the 2-item Hunger Vital Sign (Hager et al., 2010); participants who responded positively to at least one of the 2 questions were considered to live in a food insecure household. Fruit and vegetable consumption was assessed with a modified version of the Behavior Risk Factor Surveillance System Fruit and Vegetable Module Questionnaire (Lee & Moore, 2020), which excluded 100% fruit juice. The screener asks about frequency of consumption of 5 items: fruits, green leafy or lettuce salads, fried potatoes, other potatoes, and other vegetables during the previous month, with possible response options "never," "1 time last month," "2-3 times last month," "1 time per week," "2 times per week," "3-4 times per week," "5-6 times per week," "1 time per day," and "2 or more times per day." For analysis, we converted all responses into frequency of fruit or vegetable consumption per day and then categorized it into <3 times per day, ≥ 3 times per day, or missing.

This study is part of a larger project evaluating the impact and implementation equity of the Fresh Bucks program. The (blinded for review) Institutional Review Board deemed that portions of the project that solely utilize secondary data without personally identifiable information, such as this study, were not human subjects research (IRB ID: blinded for review).

Statistical analysis

We summarized the characteristics of Fresh Bucks participants using descriptive statistics, for the full sample and stratified by program (dis)continuation status. We estimated probability of program discontinuation by fitting a linear probability model in the full sample, with program discontinuation (yes/no) as the dependent variable and the following independent variables: participants' age, race/ethnicity, preferred language, program enrollment pathway, fruit and vegetable consumption, presence of children in the household, household poverty, and food security. To assess if factors associated with program discontinuation varied by preferred language, we further stratified our analysis by language preference, running linear probability models for those who preferred to speak English and those who preferred to speak a non-English language separately. These models were parametrized exactly as the one with the full sample except excluding preferred language as an independent variable. We chose to fit linear probability models given its computational and interpretation advantages over logit or probit models (avoiding nonlinear transformations or calculations of marginal effects), with the results interpreted as risk differences as opposed to ratios (Cheung, 2007). As a robustness check, we also fitted Poisson regression models with robust standard errors, with results qualitatively similar to those presented here (data available upon request). We conducted all analyses in Stata 19 and used a $p\text{-value} < 0.05$ to denote statistical significance.

Results

Table 1 presents baseline sample characteristics of Fresh Bucks participants by discontinuation status. Almost half (46%) of Fresh Bucks participants discontinued participation two years post initial program enrollment. Age composition differed by discontinuation status, with a lower proportion of older (60+ years) participants in the discontinuation (32.5%) vs. the continuation

(50.7%) group. Those in the discontinuation group had a lower proportion of households with children (61.2%) compared with the continuation group (68.0%). There were also notable differences by race/ethnicity and preferred language, with a lower proportion of NH Asian (23.8% vs. 39.9%) and Chinese speakers (8.6% vs. 22.2%) and a higher proportion of NH Black (32.4% vs. 22.1%) and English speakers (64.4% vs. 48.8%) in the discontinuation vs. continuation groups. Food insecurity prevalence was 89% for those in the discontinuation group and 86.5% in the continuation group. There were no meaningful differences in (dis)continuation status by program enrollment pathway, household poverty, and FV consumption.

Table 2 presents the results obtained from linear probability models. Across specifications, the variable with the highest magnitude of association with discontinued Fresh Bucks participation was age, with probability of discontinuation significantly decreasing as age increased. Compared to those aged 18-29 years, being aged 70+ years was associated with 26 percentage points lower probability of discontinuation in the main model ($B = -0.259$, 95% confidence interval $[CI] = -0.294, -0.223$), with similar magnitudes observed in the stratified analysis for both those who preferred to speak English and those who preferred to speak a non-English language. In the main model, race/ethnicity and preferred language were also significantly associated with Fresh Bucks program discontinuation: Self-identifying as NH Asian and reporting a preference for speaking Chinese, Spanish, or Another non-English language were associated with 4.3 ($B_{NH\ Asian} = -0.043$, 95%CI= -0.081, -0.006), 15.1 ($B_{Chinese} = -0.151$, 95%CI= -0.191, -0.112), 10.3 ($B_{Spanish} = -0.103$, 95%CI= -0.162, -0.045), and 5.9 ($B_{Another} = -0.059$, 95%CI= -0.090, -0.028) percentage points lower probability of discontinuation, respectively, compared against NH White participants and those who preferred to speak English. Conversely, self-

identifying as NH Black, compared to NH White, was associated with 5.6 percentage points higher probability of program discontinuation ($B = 0.056$, 95% CI= 0.026, 0.086).

Having a child in the household ($B_{\text{child}} = -0.038$, 95%CI= -0.062, -0.014) and a higher IPR ($B_{(100-200\% \text{ IPR})} = -0.061$, 95%CI= -0.084, -0.038; $B_{(>200\% \text{ IPR})} = -0.043$, 95%CI= -0.069, -0.018) were associated with lower probability of program discontinuation, while enrollment pathway was not associated with program discontinuation in the main model. Baseline food security status was also associated with Fresh Bucks program discontinuation in the main model. Those who reported food security at baseline had 3.6 percentage points lower probability of program discontinuation compared to those who reported food insecurity ($B = -0.036$, 95%CI= -0.064, -0.008).

In the model stratified by language preference, age, race/ethnicity, having a child in the household, and household IPR were significantly associated with program discontinuation for those who preferred to speak English, following the same direction and similar magnitude to that observed in the full sample model (**Table 2**). For those who preferred to speak a non-English language, age and food security were significantly associated with program discontinuation in the same direction and similar magnitude as the full sample model. Program enrollment pathway emerged as a significant factor associated with program discontinuation in language-stratified analysis: for those who preferred to speak a non-English language, CBO-assisted enrollment (vs. public enrollment) was associated with 7.3 percentage points *lower* probability of program discontinuation ($B = -0.073$, 95%CI= -0.105, -0.041) (**Table 2**).

Discussion

In this study, we aimed to identify factors that were associated with discontinuation of Fresh Bucks participation 2 years after the initial program launch. Further, based on previous research indicating higher program engagement among those who preferred to speak Chinese, Vietnamese, or Spanish (vs. English) (Wallace et al., unpublished results(a)), we assessed if the factors associated with program discontinuation varied by participants' preferred language. We found that 46% of initial program participants discontinued participation during the evaluated period. Age, race/ethnicity, preferred language, not having a child in the household, household poverty, and baseline food security status were associated with program discontinuation. In addition, program enrollment pathway emerged as an important correlate of program discontinuation when the sample was stratified by participant preferred language, with CBO-assisted enrollment associated with a lower probability of program discontinuation among those who preferred to speak a non-English language, when compared with public enrollment.

The finding that almost half of Fresh Bucks participants discontinued participation when prompted to re-enroll is an important one. Administrative burden, including re-certification requests, is a known determinant of discontinued participation for food and nutrition assistance programs, even when financial need is still present (Gray, 2019; Fox et al., 2023; Kenney et al., 2022). The timeline of this study corresponds with the COVID-19 pandemic, which triggered an unprecedented response from the federal government in the form of increased financial and food assistance (Patton et al., n.d.); this increase remained largely in place in Washington state until 2022-2023 (Washington State Department of Social and Health Services, n.d.), past the October 2021 request for Fresh Bucks re-application. In addition to the increased generosity of federal government assistance, local governments and non-profit organizations launched and/or expanded programs to address the increase in food insecurity triggered by the COVID-19

pandemic; an example of this is the partnership between United Way of King County and DoorDash to deliver food boxes to low-income households in King County (which includes Seattle) (United Way King County, n.d.). It is possible that having access to food via other programs and/or receiving larger-than-usual benefits from usual programs influenced the decision of people to not re-enroll in Fresh Bucks. Previous research investigating reasons why families discontinue participation in the Special Supplemental Nutrition Program for Women, Infants and Children (WIC) while still eligible have identified receipt of food from other sources (e.g., Supplemental Nutrition Assistance Program [SNAP], food banks) as an important factor (Terry et al., 2024), particularly when the other program benefits are perceived as more generous or more flexible (Weber et al., 2019).⁷ On the other hand, food inflation was still very high in the late quarter of 2021 (U.S. Bureau of Labor Statistics, n.d.), and multiple program participation was common for certain low-income households during the COVID-19 pandemic (Chaparro et al., 2025a).

The association between age and Fresh Bucks program discontinuation was the one with the highest magnitude, with probability of program discontinuation decreasing in a somewhat linear fashion as age increased. Participants aged 70+ years had 26 percentage points lower probability of program discontinuation than those aged 18-29 years. A previous study of Fresh Bucks participants found that adults 60+ years had 3.5 times the odds of redeeming >75% of their Fresh Bucks benefits (vs. <25% of the benefit), compared to 18–39-year-olds (Wallace et al., unpublished results (b)). Combined, these results suggest that older adults are highly engaged in the program. This is unlike other food assistance programs such as SNAP, for which participation rates of adults 60+ years are significantly lower than for those aged <60 years (55% vs. 88% in FY2022) (Vigil & Rahimi, 2024). It is important to note that some of the Fresh Bucks

CBO partners focus primarily on older adult services, which may partly explain their high engagement with Fresh Bucks.

We found that those who were food secure, compared to food insecure, and those who were above the poverty threshold, compared to those below the poverty threshold, had lower probability of Fresh Bucks program discontinuation. These results may seem counterintuitive, since one would anticipate that those most in need would decide to continue participating in Fresh Bucks when provided with the opportunity. Contrary to our results, Kenney et al. (2022) investigated the rate of SNAP participation discontinuation due to administrative burden among households with children in Massachusetts and found that lower-income households were disproportionately affected. On the other hand, Gray (2019) argues that different type of beneficiaries respond differently to certain kinds of administrative burden, and they identified that higher-income households with children were more likely to discontinue SNAP participation if faced with increased paperwork burden. Research on WIC participation is mixed; on the one hand, having a household income below the poverty level was associated with lower redemption of WIC foods (Anderson & Whaley, 2025) – and lower redemption of WIC foods with higher likelihood of WIC program discontinuation (Anderson et al., 2023) – among WIC participants in Southern California, whereas a lower income was associated with longer duration of WIC participation in a longitudinal analysis of national data (Anderson et al., 2022). Another explanation for our results may be that these most-in-need groups were receiving other income and food assistance in addition to Fresh Bucks which, as stated earlier, may have been more generous than usual given the COVID-19 pandemic augmentations (Washington State Department of Social and Health Services, n.d.). Receiving benefits from other programs, potentially more generous than Fresh Bucks, may have reduced their need to keep enrollment in

Fresh Bucks, particularly when faced with added administrative burden. Though it is important to note that establishing eligibility and receiving benefits from federal food assistance programs like SNAP is a very onerous process. Lastly, lower-income households may have been more likely to be priced out of Seattle, leading them to move out of the Fresh Bucks geographic eligibility area. However, eviction moratoriums and other rental assistance implemented during the COVID-19 pandemic (Rental Housing Association of WA, n.d.) may have prevented low-income households from moving out of Seattle during this time.

Race/ethnicity was associated with program discontinuation. Compared to NH White participants, we found that NH Asian participants had lower probability of program discontinuation, whereas NH Black participants had higher probability of program discontinuation. Previous research on Fresh Bucks benefit redemption reports similar findings, with NH Asian households having higher benefit redemption rates, and NH Black household lower benefit redemption rates, than NH White households (Wallace et al., unpublished results (a)). Research on WIC food benefit redemption also found lower redemption rates among NH Black families (Chaparro et al., 2025b). On the other hand, a rigorous evaluation on the impact of Fresh Bucks participation found that NH Black participants increased their FV consumption because of Fresh Bucks participation (Knox et al., 2025). Thus, to inform retention efforts, future research should seek to understand why NH Black participants were more likely to discontinue participation.

Having a preference for speaking Chinese, Spanish, or another non-English language, vs. English, was associated with lower probability of program discontinuation. Similar findings were reported previously in terms of Fresh Bucks benefit redemption (Wallace et al., unpublished results (a)). It is important to note that Fresh Bucks partners with a network of CBOs for

recruitment and engagement of historically underserved populations, including those whose preferred language is not English. Some of these CBOs focus on specific demographic groups organized by culture/language and, thus, provide culturally and linguistically tailored support to certain Fresh Bucks participants. In fact, when the sample was stratified by language preference (English vs. non-English), CBO-assisted enrollment emerged as a significant factor associated with lower probability of program discontinuation among those who prefer to speak a non-English language. Previous qualitative studies investigating barriers and facilitators of implementation of Food Is Medicine programs in clinical settings (e.g., produce prescription programs, medically tailored meals) identified community engagement/partnership as an important facilitator (Houghtaling et al., 2025; Houghtaling et al., 2024; McWhorter et al., 2023). Furthermore, a process evaluation of *Health Bucks*, a farmers' market coupon incentive program in New York City, reported that using CBOs for coupon distribution may have increased awareness of the program as well as facilitated nutrition education activities, which were carried out by CBOs (Payne et al., 2013). Other public health research has also highlighted the importance of partnering with CBOs to recruit and retain underserved populations in health promoting programs (Steinman et al., 2022; O'Mara-Eves et al., 2015).

This study has strengths and limitations. Strengths include the availability of longitudinal administrative program data, which allowed us to identify those who discontinued participation when requested to re-apply. The large sample size and diversity of Fresh Bucks participants in terms of sociodemographic characteristics are also strengths, as we were sufficiently powered to assess how probability of program discontinuation varied by these characteristics. Another strength is the use of food security (Hager et al., 2010) and fruit and vegetable consumption (Lee & Moore, 2020) measurement tools that have been previously validated. In terms of limitations,

given our data we are unable to ascertain if program discontinuation was due to loss of eligibility (i.e., income changes, relocation outside the city of Seattle) vs. personal choice, administrative burden, time constraints, or communication barriers. The overlap of our study dates with the COVID-19 pandemic is another limitation; it is possible that the factors identified as associated with program discontinuation would be different if assessed at a different time. Lastly, our study is based on one city-wide fruit and vegetable incentive program, so our results may not be generalizable to other incentive programs with different target populations, eligibility criteria, benefit amounts, or incentive scheme. It is important to note, though, that most of our results are similar to those previously reported across a variety of food and nutrition assistance programs.

In conclusion, Fresh Bucks retained a substantial share of the city's most socially and economically vulnerable households, especially older adults and those who preferred to speak a non-English language. Age, race/ethnicity, language, and income were key factors associated with discontinuation, with CBO-assisted enrollment emerging as a critical facilitator of retention among non-English-speaking households. These findings highlight the importance of using community engagement strategies to maximize the long-term impact of healthy food voucher programs for historically underserved populations.

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Table 1: Baseline sample characteristics of Fresh Bucks participants by Fresh Bucks

(dis)continuation status (n=12,813)

	Continued participation N (%)	Discontinued participation N (%)	Total sample N (%)
N	6,864	5,949	12,813
Age (years)			
18-29	513 (7.5)	990 (16.9)	1,503 (11.9)
30-39	1,061 (15.6)	1,294 (22.0)	2,355 (18.6)
40-49	942 (13.9)	940 (16.0)	1,882 (14.8)
50-59	840 (12.4)	741 (12.6)	1,581 (12.5)
60-69	1,350 (19.9)	853 (14.5)	2,203 (17.4)
70+	2,095 (30.8)	1,055 (18.0)	3,150 (24.9)
Race/ethnicity			
NH Asian	2,647 (39.9)	1,357 (23.8)	4,004 (32.5)
NH Black	1,465 (22.1)	1,843 (32.4)	3,308 (26.8)
Hispanic/Latino	750 (11.3)	667 (11.7)	1,417 (11.5)
NH White	1,232 (18.6)	1,153 (20.2)	2,385 (19.3)
NH Other	546 (8.2)	675 (11.9)	1,221 (9.9)
Preferred language			
Chinese	1,526 (22.2)	514 (8.6)	2,040 (15.9)
English	3,351 (48.8)	3,830 (64.4)	7,181 (56.0)
Spanish	610 (8.9)	470 (7.9)	1,080 (8.4)
Vietnamese	538 (7.8)	357 (6.0)	895 (7.0)
Another non-English	839 (12.2)	778 (13.1)	1,617 (12.6)
Child in the household			
Yes	4,605 (68.0)	3,598 (61.2)	8,203 (64.9)
No	2,165 (32.0)	2,278 (38.8)	4,443 (35.1)
Household IPR			
<100%	4,072 (59.3)	3,563 (59.9)	7,635 (59.6)
100-200%	1,634 (23.8)	1,280 (21.5)	2,914 (22.7)
>200%	1,158 (16.9)	1,106 (18.6)	2,264 (17.7)
Enrollment pathway			
Public	4,068 (59.27)	3,413 (57.37)	7,481 (58.39)
CBO-assisted	2,796 (40.73)	2,536 (42.63)	5,332 (41.61)
Food security status¹			
Food secure	839 (13.5)	594 (10.9)	1,433 (12.3)
Food insecure	5,357 (86.5)	4,873 (89.1)	10,230 (87.7)
Fruit and vegetable consumption²			
<3 times per day	5,334 (77.7)	4,773 (80.2)	10,107 (78.8)
≥3 times per day	1,216 (17.7)	911 (15.31)	2,127 (16.6)
Missing	314 (4.6)	265 (4.5)	579 (4.5)

Abbreviations: CBO = community-based organization; NH = non-Hispanic; IPR = income-to-poverty ratio¹Measured with the 2-item Hunger Vital Sign;⁷ participants who responded positively to at least one of the 2 questions were considered to live in a food insecure household.²Measured with a modified version of the Behavior Risk Factor Surveillance System Fruit and Vegetable Module Questionnaire,⁸ excluding 100% fruit juice.

Table 2: Results from linear probability models assessing predictors of Fresh Bucks program discontinuation, main model and stratified by preferred language

	Main model (n=11,098)		Preferred English (n=6,265)		Stratified models Preferred non-English language (n=4,706)	
	B	95% CI	B	95% CI	B	95% CI
Age (years)						
18-29	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
30-39	-0.092**	-0.126, -0.059	-0.070**	-0.109, -0.031	-0.146**	-0.217, -0.076
40-49	-0.133**	-0.167, -0.098	-0.117**	-0.159, -0.076	-0.165**	-0.236, -0.094
50-59	-0.177**	-0.214, -0.141	-0.185**	-0.228, -0.142	-0.172**	-0.247, -0.097
60-69	-0.237**	-0.272, -0.202	-0.249**	-0.292, -0.207	-0.231**	-0.306, -0.157
70+	-0.259**	-0.294, -0.223	-0.268**	-0.313, -0.224	-0.256**	-0.330, -0.182
Race/ethnicity						
NH Asian	-0.043*	-0.081, -0.006	-0.070**	-0.111, -0.028	-0.079	-0.173, 0.015
NH Black	0.056**	0.026, 0.086	0.040*	0.008, 0.072	0.100	-0.002, 0.201
Hispanic/Latino	0.024	-0.030, 0.079	0.026	-0.030, 0.083	0.039	-0.065, 0.142
NH White	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
NH Other	0.052**	0.016, 0.089	0.031	-0.008, 0.070	0.109	-0.022, 0.240
Preferred language						
Chinese	-0.151**	-0.191, -0.112				
English	Ref.	Ref.				
Spanish	-0.103**	-0.162, -0.045				
Vietnamese	-0.044	-0.091, 0.004				
Another non-English	-0.059**	-0.090, -0.028				
Child in the household						
Yes	-0.038**	-0.062, -0.014	-0.038*	-0.068, -0.008	-0.028	-0.070, 0.014
No	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Household IPR						
<100%	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
100-200%	-0.061**	-0.084, -0.038	-0.076**	-0.105, -0.046	-0.018	-0.056, 0.020
>200%	-0.043*	-0.069, -0.018	-0.050**	-0.081, -0.019	-0.042	-0.090, 0.006
Enrollment pathway						
Public	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
CBO-assisted	-0.015	-0.035, 0.005	0.025	-0.001, 0.051	-0.073**	-0.105, -0.041
Food security status¹						

Food insecure	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
Food secure	-0.036*	-0.064, -0.008	-0.032	-0.071, 0.008	-0.047*	-0.087, -0.008
Fruit and vegetable consumption²						
<3 times per day	Ref.	Ref.	Ref.	Ref.	Ref.	Ref.
≥3 times per day	0.006	-0.018, 0.030	-0.001	-0.035, 0.032	0.010	-0.024, 0.044
Missing	0.100	-0.032, 0.232	0.108	-0.052, 0.268	0.086	-0.160, 0.333

Abbreviations: CBO = community-based organization; NH = non-Hispanic; IPR = income-to-poverty ratio

¹Measured with the 2-item Hunger Vital Sign;⁷ participants who responded positively to at least one of the 2 questions were considered to live in a food insecure household.

²Measured with a modified version of the Behavior Risk Factor Surveillance System Fruit and Vegetable Module Questionnaire,⁸ excluding 100% fruit juice.

* p < 0.05, ** p < 0.01